Received Order Analysis Engine (ROAE)

Analysis of the King Wen sequence including observations by Terence McKenna

Hexagram reference table

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Hexagram reference table
Each hexagram is a stack of 6 lines, either solid (1/yang) or broken (0/yin).
The 'binary' column encodes these 6 lines as bits (bottom line = rightmost bit).
Each hexagram is also composed of two 3-line trigrams: an upper and a lower.
There are 8 possible trigrams (Heaven, Earth, Thunder, Water, Mountain, Wind,
Fire, Lake), giving 8x8 = 64 possible hexagrams.
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Pos | Hex | Binary | Upper          | Lower          | Name
----|-----|--------|----------------|----------------|----
01  | ䷀   | 111111 | Qian Heaven    | Qian Heaven    | The Creative
02  | ䷁   | 000000 | Kun  Earth     | Kun  Earth     | The Receptive
03  | ䷂   | 010001 | Kan  Water     | Zhen Thunder   | Difficulty at the Beginning
04  | ䷃   | 100010 | Gen  Mountain  | Kan  Water     | Youthful Folly
05  | ䷄   | 010111 | Kan  Water     | Qian Heaven    | Waiting
06  | ䷅   | 111010 | Qian Heaven    | Kan  Water     | Conflict
07  | ䷆   | 000010 | Kun  Earth     | Kan  Water     | The Army
08  | ䷇   | 010000 | Kan  Water     | Kun  Earth     | Holding Together
09  | ䷈   | 110111 | Xun  Wind      | Qian Heaven    | Small Taming
10  | ䷉   | 111011 | Qian Heaven    | Dui  Lake      | Treading
11  | ䷊   | 000111 | Kun  Earth     | Qian Heaven    | Peace
12  | ䷋   | 111000 | Qian Heaven    | Kun  Earth     | Standstill
13  | ䷌   | 111101 | Qian Heaven    | Li   Fire      | Fellowship
14  | ䷍   | 101111 | Li   Fire      | Qian Heaven    | Great Possession
15  | ䷎   | 000100 | Kun  Earth     | Gen  Mountain  | Modesty
16  | ䷏   | 001000 | Zhen Thunder   | Kun  Earth     | Enthusiasm
17  | ䷐   | 011001 | Dui  Lake      | Zhen Thunder   | Following
18  | ䷑   | 100110 | Gen  Mountain  | Xun  Wind      | Work on the Decayed
19  | ䷒   | 000011 | Kun  Earth     | Dui  Lake      | Approach
20  | ䷓   | 110000 | Xun  Wind      | Kun  Earth     | Contemplation
21  | ䷔   | 101001 | Li   Fire      | Zhen Thunder   | Biting Through
22  | ䷕   | 100101 | Gen  Mountain  | Li   Fire      | Grace
23  | ䷖   | 100000 | Gen  Mountain  | Kun  Earth     | Splitting Apart
24  | ䷗   | 000001 | Kun  Earth     | Zhen Thunder   | Return
25  | ䷘   | 111001 | Qian Heaven    | Zhen Thunder   | Innocence
26  | ䷙   | 100111 | Gen  Mountain  | Qian Heaven    | Great Taming
27  | ䷚   | 100001 | Gen  Mountain  | Zhen Thunder   | Nourishment
28  | ䷛   | 011110 | Dui  Lake      | Xun  Wind      | Great Preponderance
29  | ䷜   | 010010 | Kan  Water     | Kan  Water     | The Abysmal
30  | ䷝   | 101101 | Li   Fire      | Li   Fire      | The Clinging
31  | ䷞   | 011100 | Dui  Lake      | Gen  Mountain  | Influence
32  | ䷟   | 001110 | Zhen Thunder   | Xun  Wind      | Duration
33  | ䷠   | 111100 | Qian Heaven    | Gen  Mountain  | Retreat
34  | ䷡   | 001111 | Zhen Thunder   | Qian Heaven    | Great Power
35  | ䷢   | 101000 | Li   Fire      | Kun  Earth     | Progress
36  | ䷣   | 000101 | Kun  Earth     | Li   Fire      | Darkening of the Light
37  | ䷤   | 110101 | Xun  Wind      | Li   Fire      | The Family
38  | ䷥   | 101011 | Li   Fire      | Dui  Lake      | Opposition
39  | ䷦   | 010100 | Kan  Water     | Gen  Mountain  | Obstruction
40  | ䷧   | 001010 | Zhen Thunder   | Kan  Water     | Deliverance
41  | ䷨   | 100011 | Gen  Mountain  | Dui  Lake      | Decrease
42  | ䷩   | 110001 | Xun  Wind      | Zhen Thunder   | Increase
43  | ䷪   | 011111 | Dui  Lake      | Qian Heaven    | Breakthrough
44  | ䷫   | 111110 | Qian Heaven    | Xun  Wind      | Coming to Meet
45  | ䷬   | 011000 | Dui  Lake      | Kun  Earth     | Gathering Together
46  | ䷭   | 000110 | Kun  Earth     | Xun  Wind      | Pushing Upward
47  | ䷮   | 011010 | Dui  Lake      | Kan  Water     | Oppression
48  | ䷯   | 010110 | Kan  Water     | Xun  Wind      | The Well
49  | ䷰   | 011101 | Dui  Lake      | Li   Fire      | Revolution
50  | ䷱   | 101110 | Li   Fire      | Xun  Wind      | The Cauldron
51  | ䷲   | 001001 | Zhen Thunder   | Zhen Thunder   | The Arousing
52  | ䷳   | 100100 | Gen  Mountain  | Gen  Mountain  | Keeping Still
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53  | ䷴   | 110100 | Xun  Wind      | Gen  Mountain  | Development
54  | ䷵   | 001011 | Zhen Thunder   | Dui  Lake      | The Marrying Maiden
55  | ䷶   | 001101 | Zhen Thunder   | Li   Fire      | Abundance
56  | ䷷   | 101100 | Li   Fire      | Gen  Mountain  | The Wanderer
57  | ䷸   | 110110 | Xun  Wind      | Xun  Wind      | The Gentle
58  | ䷹   | 011011 | Dui  Lake      | Dui  Lake      | The Joyous
59  | ䷺   | 110010 | Xun  Wind      | Kan  Water     | Dispersion
60  | ䷻   | 010011 | Kan  Water     | Dui  Lake      | Limitation
61  | ䷼   | 110011 | Xun  Wind      | Dui  Lake      | Inner Truth
62  | ䷽   | 001100 | Zhen Thunder   | Gen  Mountain  | Small Preponderance
63  | ䷾   | 010101 | Kan  Water     | Li   Fire      | After Completion
64  | ䷿   | 101010 | Li   Fire      | Kan  Water     | Before Completion
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Reverse vs. Inverse pair analysis

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Reverse vs. Inverse pair analysis
The 64 hexagrams are traditionally grouped into 32 pairs (1-2, 3-4, ..., 63-64).
For each pair, we test two relationships:
  Reverse: flip the hexagram upside down (read the lines top-to-bottom instead of
           bottom-to-top). If the second hexagram upside-down equals the first,
           they are a reverse pair.
  Inverse: toggle every line (solid becomes broken, broken becomes solid).
           If toggling all lines of one hexagram gives the other, they are inverse.
Key finding: EVERY pair in the King Wen sequence is one or the other — none are
unrelated. This perfect pairing structure is unlikely to occur by chance.
---
01 ䷀ Inverse 02 ䷁
03 ䷂ Reverse 04 ䷃
05 ䷄ Reverse 06 ䷅
07 ䷆ Reverse 08 ䷇
09 ䷈ Reverse 10 ䷉
11 ䷊ Reverse 12 ䷋
13 ䷌ Reverse 14 ䷍
15 ䷎ Reverse 16 ䷏
17 ䷐ Reverse 18 ䷑
19 ䷒ Reverse 20 ䷓
21 ䷔ Reverse 22 ䷕
23 ䷖ Reverse 24 ䷗
25 ䷘ Reverse 26 ䷙
27 ䷚ Inverse 28 ䷛
29 ䷜ Inverse 30 ䷝
31 ䷞ Reverse 32 ䷟
33 ䷠ Reverse 34 ䷡
35 ䷢ Reverse 36 ䷣
37 ䷤ Reverse 38 ䷥
39 ䷦ Reverse 40 ䷧
41 ䷨ Reverse 42 ䷩
43 ䷪ Reverse 44 ䷫
45 ䷬ Reverse 46 ䷭
47 ䷮ Reverse 48 ䷯
49 ䷰ Reverse 50 ䷱
51 ䷲ Reverse 52 ䷳
53 ䷴ Reverse 54 ䷵
55 ䷶ Reverse 56 ䷷
57 ䷸ Reverse 58 ䷹
59 ䷺ Reverse 60 ䷻
61 ䷼ Inverse 62 ䷽
63 ䷾ Reverse 64 ䷿
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Reverse count pairs: 28/32 (87.5%)
Inverse count pairs: 04/32 (12.5%)
Neither count pairs: 00/32 (0.0%)
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First order of difference (the 'wave')

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For each consecutive pair of hexagrams, count how many of the 6 lines change.
This produces a sequence of values from 0 (identical) to 6 (every line flipped).
The resulting 'wave' is central to analysis of the King Wen sequence.
Key observation (McKenna, The Invisible Landscape, 1975): the value 5 never appears — no consecutive hexagrams
differ by exactly 5 lines. The value 0 also never appears (no duplicates).
---
01 ䷀ to 02 ䷁ difference 6 ******
02 ䷁ to 03 ䷂ difference 2 **
03 ䷂ to 04 ䷃ difference 4 ****
04 ䷃ to 05 ䷄ difference 4 ****
05 ䷄ to 06 ䷅ difference 4 ****
06 ䷅ to 07 ䷆ difference 3 ***
07 ䷆ to 08 ䷇ difference 2 **
08 ䷇ to 09 ䷈ difference 4 ****
09 ䷈ to 10 ䷉ difference 2 **
10 ䷉ to 11 ䷊ difference 4 ****
11 ䷊ to 12 ䷋ difference 6 ******
12 ䷋ to 13 ䷌ difference 2 **
13 ䷌ to 14 ䷍ difference 2 **
14 ䷍ to 15 ䷎ difference 4 ****
15 ䷎ to 16 ䷏ difference 2 **
16 ䷏ to 17 ䷐ difference 2 **
17 ䷐ to 18 ䷑ difference 6 ******
18 ䷑ to 19 ䷒ difference 3 ***
19 ䷒ to 20 ䷓ difference 4 ****
20 ䷓ to 21 ䷔ difference 3 ***
21 ䷔ to 22 ䷕ difference 2 **
22 ䷕ to 23 ䷖ difference 2 **
23 ䷖ to 24 ䷗ difference 2 **
24 ䷗ to 25 ䷘ difference 3 ***
25 ䷘ to 26 ䷙ difference 4 ****
26 ䷙ to 27 ䷚ difference 2 **
27 ䷚ to 28 ䷛ difference 6 ******
28 ䷛ to 29 ䷜ difference 2 **
29 ䷜ to 30 ䷝ difference 6 ******
30 ䷝ to 31 ䷞ difference 3 ***
31 ䷞ to 32 ䷟ difference 2 **
32 ䷟ to 33 ䷠ difference 3 ***
33 ䷠ to 34 ䷡ difference 4 ****
34 ䷡ to 35 ䷢ difference 4 ****
35 ䷢ to 36 ䷣ difference 4 ****
36 ䷣ to 37 ䷤ difference 2 **
37 ䷤ to 38 ䷥ difference 4 ****
38 ䷥ to 39 ䷦ difference 6 ******
39 ䷦ to 40 ䷧ difference 4 ****
40 ䷧ to 41 ䷨ difference 3 ***
41 ䷨ to 42 ䷩ difference 2 **
42 ䷩ to 43 ䷪ difference 4 ****
43 ䷪ to 44 ䷫ difference 2 **
44 ䷫ to 45 ䷬ difference 3 ***
45 ䷬ to 46 ䷭ difference 4 ****
46 ䷭ to 47 ䷮ difference 3 ***
47 ䷮ to 48 ䷯ difference 2 **
48 ䷯ to 49 ䷰ difference 3 ***
49 ䷰ to 50 ䷱ difference 4 ****
50 ䷱ to 51 ䷲ difference 4 ****
51 ䷲ to 52 ䷳ difference 4 ****
52 ䷳ to 53 ䷴ difference 1 *
53 ䷴ to 54 ䷵ difference 6 ******
54 ䷵ to 55 ䷶ difference 2 **
55 ䷶ to 56 ䷷ difference 2 **
56 ䷷ to 57 ䷸ difference 3 ***
57 ䷸ to 58 ䷹ difference 4 ****
58 ䷹ to 59 ䷺ difference 3 ***
59 ䷺ to 60 ䷻ difference 2 **
60 ䷻ to 61 ䷼ difference 1 *
61 ䷼ to 62 ䷽ difference 6 ******
62 ䷽ to 63 ䷾ difference 3 ***
63 ䷾ to 64 ䷿ difference 6 ******
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0 line changes total 0
1 line changes total 2
2 line changes total 20
3 line changes total 13
4 line changes total 19
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5 line changes total 0
6 line changes total 9

Spark line: █▂▅▅▅▃▂▅▂▅█▂▂▅▂▂█▂▅▃▂▂▂▃▅▂█▂█▃▂▃▅▅▅▂▅█▅▃▂▅▂▃▅▃▂▃▅▅▅▁█▂▂▃▅▃▂▁█▅█

--- Wald-Wolfowitz runs test ---
Tests whether the wave values are randomly ordered (vs. clustered or alternating).
Values split at median (3): 28 above, 22 below (13 excluded ties)
Observed runs: 33, expected: 25.6, Z = +2.13
Significant at 95% level (p = 0.033): the wave shows alternation.
Note: does not survive Bonferroni correction for 28 tests (threshold p < 0.0018).
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Difference wave bar chart

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Difference wave bar chart
A visual representation of the first-order difference wave. Each column shows
how many lines changed in that transition. The horizontal axis is the position
in the King Wen sequence (1-63), the vertical axis is the number of lines
changed (1-6). This makes the wave pattern easier to see than a list of numbers.
---
  6 |#         #     #         # #        #              #       # #|
  5 |#         #     #         # #        #              #       # #|
  4 |# ###  # ##  #  # #     # # #   ### ###  #  #   ### #   #   # #|
  3 |# #### # ##  #  ####   ## # ## #### #### # ### #### #  ###  ###|
  2 |################################################### ####### ###|
  1 |###############################################################|
    +---------------------------------------------------------------+
              1         2         3         4         5         6
              0         0         0         0         0         0
```

Trigram analysis

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Trigram analysis
Each hexagram is built from two trigrams (3-line figures): an upper and a lower.
There are 8 possible trigrams. This section counts how often each trigram appears
in each position, checks how often the upper or lower trigram changes between
consecutive hexagrams, and shows the full 8x8 transition matrix (which trigram
follows which). A uniform distribution (8 of each) confirms that all 64 possible
upper/lower combinations are used exactly once.
---
Trigram             Upper  Lower  Total
Kun  Earth            8      8     16
Zhen Thunder          8      8     16
Kan  Water            8      8     16
Dui  Lake             8      8     16
Gen  Mountain         8      8     16
Li   Fire             8      8     16
Xun  Wind             8      8     16
Qian Heaven           8      8     16

--- Consecutive trigram transitions ---
Upper trigram changes: 59/63 transitions (93.7%)
Lower trigram changes: 58/63 transitions (92.1%)

--- Upper trigram transition matrix ---
Rows = from, Columns = to
          Kun   Zhen  Kan   Dui   Gen   Li    Xun   Qian
Kun       0     1     2     1     0     0     2     2
Zhen      0     1     1     1     2     2     0     1
Kan       0     1     0     1     1     2     2     1
Dui       1     1     2     0     1     1     1     1
Gen       2     0     1     1     2     0     2     0
Li        2     1     1     1     1     0     1     0
Xun       0     2     1     2     0     2     0     1
Qian      3     1     0     1     1     1     0     1
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--- Lower trigram transition matrix ---
Rows = from, Columns = to
          Kun   Zhen  Kan   Dui   Gen   Li    Xun   Qian
Kun       0     4     0     0     0     2     1     1
Zhen      0     1     1     0     1     1     2     2
Kan       1     0     1     2     0     1     1     1
Dui       1     1     1     1     2     1     0     1
Gen       1     0     1     1     1     1     2     1
Li        1     0     1     1     2     1     1     1
Xun       1     1     2     2     1     1     0     0
Qian      3     1     1     1     1     0     1     0

--- Trigram change rates vs null (pair-preserving permutation test) ---
Null: 10,000 orderings preserving the pair structure (permute pairs + flip orientations),
the correct baseline per CRITIQUE.md. Statistic: upper/lower trigram change counts.
KW upper changes 59/63: percentile 47.3 (fraction of null <= KW)
KW lower changes 58/63: percentile 27.3

--- Pure (doubled-trigram) hexagram placement ---
The 8 pure hexagrams (upper == lower trigram). Lai Zhide (1525-1604, via Schulz 1982)
observed kan/li doubles closing both Classics; measured here against the same null.
Pure hexagram positions (1-based): [1, 2, 29, 30, 51, 52, 57, 58]
Pure hexagrams at Classic ends (positions 1,2,29,30,63,64): 4 of a possible 6
Null P(>= KW's 4) = 0.0336  (pair-preserving null; note KW's C4 fixes 1,2 by
definition, so interpret against the constrained baseline)

--- Nuclear trigram structure ---
Nuclear hexagram = lines 2-4 (lower nuclear) + 3-5 (upper nuclear). Classical fact
(commentary tradition): iterating the nuclear map sends all 64 hexagrams to 16, then to
the 4 fixed points 63, 0, and the two alternators 21(010101b rendering varies), 42.
Distinct after 1 application: 16; after 2: 4; after 3: 4 -> [0, 21, 42, 63]
KW nuclear-hexagram changes along the sequence: 58/63

--- Jing Fang Eight-Palaces rank correlation ---
Spearman rank correlation between KW positions and Jing Fang's palace ordering
(the trigram-generated classical alternative; ordering as in solve.c --null-historical).
Spearman rho = 0.1384; null P(|rho| >= observed) = 0.1170 (pair-preserving null)

--- Symmetry group vs the trigram decomposition (ROAE SYMMETRY_SEARCH theorem) ---
The C1-C5 symmetry group B3 (order 48) acts on LINE positions; reversal maps the upper
trigram to the reversed lower trigram, so most of the group does NOT preserve the
upper/lower split. Preserving elements = permutations acting within {0,1,2} and {3,4,5}
blocks (or swapping them wholesale) that also centralize rev.
Split-respecting subgroup order: 12 of 48 (computed over the rev-centralizer)

--- Methodological note ---
With ~1 expected observation per cell, no goodness-of-fit test (e.g.,
chi-square) has sufficient power to detect deviations from uniform
transitions. The matrices are descriptive only.
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Nuclear hexagram analysis

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Nuclear hexagram analysis
Every hexagram contains a hidden 'nuclear' hexagram inside it, formed by
extracting the 4 middle lines (lines 2-3-4-5). Lines 2-3-4 form the lower
nuclear trigram, and lines 3-4-5 form the upper nuclear trigram (sharing lines
3 and 4). This creates a chain: each hexagram points to its nuclear hexagram,
which points to its own nuclear hexagram, and so on until the chain loops.
These chains reveal deep structural relationships between hexagrams that are
not obvious from the surface arrangement.
---
01 ䷀ nuclear: 01 ䷀  chain: 01 ䷀ -> 01 ䷀
02 ䷁ nuclear: 02 ䷁  chain: 02 ䷁ -> 02 ䷁
03 ䷂ nuclear: 23 ䷖  chain: 03 ䷂ -> 23 ䷖ -> 02 ䷁ -> 02 ䷁
04 ䷃ nuclear: 24 ䷗  chain: 04 ䷃ -> 24 ䷗ -> 02 ䷁ -> 02 ䷁
05 ䷄ nuclear: 38 ䷥  chain: 05 ䷄ -> 38 ䷥ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
06 ䷅ nuclear: 37 ䷤  chain: 06 ䷅ -> 37 ䷤ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
07 ䷆ nuclear: 24 ䷗  chain: 07 ䷆ -> 24 ䷗ -> 02 ䷁ -> 02 ䷁
08 ䷇ nuclear: 23 ䷖  chain: 08 ䷇ -> 23 ䷖ -> 02 ䷁ -> 02 ䷁
09 ䷈ nuclear: 38 ䷥  chain: 09 ䷈ -> 38 ䷥ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
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10 ䷉ nuclear: 37 ䷤  chain: 10 ䷉ -> 37 ䷤ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
11 ䷊ nuclear: 54 ䷵  chain: 11 ䷊ -> 54 ䷵ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
12 ䷋ nuclear: 53 ䷴  chain: 12 ䷋ -> 53 ䷴ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
13 ䷌ nuclear: 44 ䷫  chain: 13 ䷌ -> 44 ䷫ -> 01 ䷀ -> 01 ䷀
14 ䷍ nuclear: 43 ䷪  chain: 14 ䷍ -> 43 ䷪ -> 01 ䷀ -> 01 ䷀
15 ䷎ nuclear: 40 ䷧  chain: 15 ䷎ -> 40 ䷧ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
16 ䷏ nuclear: 39 ䷦  chain: 16 ䷏ -> 39 ䷦ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
17 ䷐ nuclear: 53 ䷴  chain: 17 ䷐ -> 53 ䷴ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
18 ䷑ nuclear: 54 ䷵  chain: 18 ䷑ -> 54 ䷵ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
19 ䷒ nuclear: 24 ䷗  chain: 19 ䷒ -> 24 ䷗ -> 02 ䷁ -> 02 ䷁
20 ䷓ nuclear: 23 ䷖  chain: 20 ䷓ -> 23 ䷖ -> 02 ䷁ -> 02 ䷁
21 ䷔ nuclear: 39 ䷦  chain: 21 ䷔ -> 39 ䷦ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
22 ䷕ nuclear: 40 ䷧  chain: 22 ䷕ -> 40 ䷧ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
23 ䷖ nuclear: 02 ䷁  chain: 23 ䷖ -> 02 ䷁ -> 02 ䷁
24 ䷗ nuclear: 02 ䷁  chain: 24 ䷗ -> 02 ䷁ -> 02 ䷁
25 ䷘ nuclear: 53 ䷴  chain: 25 ䷘ -> 53 ䷴ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
26 ䷙ nuclear: 54 ䷵  chain: 26 ䷙ -> 54 ䷵ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
27 ䷚ nuclear: 02 ䷁  chain: 27 ䷚ -> 02 ䷁ -> 02 ䷁
28 ䷛ nuclear: 01 ䷀  chain: 28 ䷛ -> 01 ䷀ -> 01 ䷀
29 ䷜ nuclear: 27 ䷚  chain: 29 ䷜ -> 27 ䷚ -> 02 ䷁ -> 02 ䷁
30 ䷝ nuclear: 28 ䷛  chain: 30 ䷝ -> 28 ䷛ -> 01 ䷀ -> 01 ䷀
31 ䷞ nuclear: 44 ䷫  chain: 31 ䷞ -> 44 ䷫ -> 01 ䷀ -> 01 ䷀
32 ䷟ nuclear: 43 ䷪  chain: 32 ䷟ -> 43 ䷪ -> 01 ䷀ -> 01 ䷀
33 ䷠ nuclear: 44 ䷫  chain: 33 ䷠ -> 44 ䷫ -> 01 ䷀ -> 01 ䷀
34 ䷡ nuclear: 43 ䷪  chain: 34 ䷡ -> 43 ䷪ -> 01 ䷀ -> 01 ䷀
35 ䷢ nuclear: 39 ䷦  chain: 35 ䷢ -> 39 ䷦ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
36 ䷣ nuclear: 40 ䷧  chain: 36 ䷣ -> 40 ䷧ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
37 ䷤ nuclear: 64 ䷿  chain: 37 ䷤ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
38 ䷥ nuclear: 63 ䷾  chain: 38 ䷥ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
39 ䷦ nuclear: 64 ䷿  chain: 39 ䷦ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
40 ䷧ nuclear: 63 ䷾  chain: 40 ䷧ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
41 ䷨ nuclear: 24 ䷗  chain: 41 ䷨ -> 24 ䷗ -> 02 ䷁ -> 02 ䷁
42 ䷩ nuclear: 23 ䷖  chain: 42 ䷩ -> 23 ䷖ -> 02 ䷁ -> 02 ䷁
43 ䷪ nuclear: 01 ䷀  chain: 43 ䷪ -> 01 ䷀ -> 01 ䷀
44 ䷫ nuclear: 01 ䷀  chain: 44 ䷫ -> 01 ䷀ -> 01 ䷀
45 ䷬ nuclear: 53 ䷴  chain: 45 ䷬ -> 53 ䷴ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
46 ䷭ nuclear: 54 ䷵  chain: 46 ䷭ -> 54 ䷵ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
47 ䷮ nuclear: 37 ䷤  chain: 47 ䷮ -> 37 ䷤ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
48 ䷯ nuclear: 38 ䷥  chain: 48 ䷯ -> 38 ䷥ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
49 ䷰ nuclear: 44 ䷫  chain: 49 ䷰ -> 44 ䷫ -> 01 ䷀ -> 01 ䷀
50 ䷱ nuclear: 43 ䷪  chain: 50 ䷱ -> 43 ䷪ -> 01 ䷀ -> 01 ䷀
51 ䷲ nuclear: 39 ䷦  chain: 51 ䷲ -> 39 ䷦ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
52 ䷳ nuclear: 40 ䷧  chain: 52 ䷳ -> 40 ䷧ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
53 ䷴ nuclear: 64 ䷿  chain: 53 ䷴ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
54 ䷵ nuclear: 63 ䷾  chain: 54 ䷵ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
55 ䷶ nuclear: 28 ䷛  chain: 55 ䷶ -> 28 ䷛ -> 01 ䷀ -> 01 ䷀
56 ䷷ nuclear: 28 ䷛  chain: 56 ䷷ -> 28 ䷛ -> 01 ䷀ -> 01 ䷀
57 ䷸ nuclear: 38 ䷥  chain: 57 ䷸ -> 38 ䷥ -> 63 ䷾ -> 64 ䷿ -> 63 ䷾
58 ䷹ nuclear: 37 ䷤  chain: 58 ䷹ -> 37 ䷤ -> 64 ䷿ -> 63 ䷾ -> 64 ䷿
59 ䷺ nuclear: 27 ䷚  chain: 59 ䷺ -> 27 ䷚ -> 02 ䷁ -> 02 ䷁
60 ䷻ nuclear: 27 ䷚  chain: 60 ䷻ -> 27 ䷚ -> 02 ䷁ -> 02 ䷁
61 ䷼ nuclear: 27 ䷚  chain: 61 ䷼ -> 27 ䷚ -> 02 ䷁ -> 02 ䷁
62 ䷽ nuclear: 28 ䷛  chain: 62 ䷽ -> 28 ䷛ -> 01 ䷀ -> 01 ䷀
63 ䷾ nuclear: 64 ䷿  chain: 63 ䷾ -> 64 ䷿ -> 63 ䷾
64 ䷿ nuclear: 63 ䷾  chain: 64 ䷿ -> 63 ䷾ -> 64 ䷿

--- Nuclear hexagram frequency ---
How often each hexagram appears as another's nuclear hexagram
01 ䷀ The Creative                    appears 4 times
02 ䷁ The Receptive                   appears 4 times
23 ䷖ Splitting Apart                 appears 4 times
24 ䷗ Return                          appears 4 times
27 ䷚ Nourishment                     appears 4 times
28 ䷛ Great Preponderance             appears 4 times
37 ䷤ The Family                      appears 4 times
38 ䷥ Opposition                      appears 4 times
39 ䷦ Obstruction                     appears 4 times
40 ䷧ Deliverance                     appears 4 times
43 ䷪ Breakthrough                    appears 4 times
44 ䷫ Coming to Meet                  appears 4 times
53 ䷴ Development                     appears 4 times
54 ䷵ The Marrying Maiden             appears 4 times
63 ䷾ After Completion                appears 4 times
64 ䷿ Before Completion               appears 4 times
```

```
--- Null model note ---
Nuclear hexagram derivation is a fixed function of binary value (lines 2-5),
independent of the King Wen ordering. The chain structure, cycle lengths,
and frequency distribution above are identical for ANY ordering of the 64
hexagrams. These properties reflect the 6-bit binary system, not the
King Wen sequence specifically.
```

Line change positional analysis

```
---
Line change positional analysis
Each hexagram has 6 lines (positions 1-6, bottom to top). When consecutive
hexagrams differ, which specific lines are changing? If the King Wen sequence
were random, each line position would change about 50% of the time. Significant
deviation from 50% would suggest the ordering favors changing certain positions.
The co-change matrix shows which pairs of lines tend to flip together in the
same transition — revealing correlated line movements.
---
Line 1: 37/63 ( 58.7%)  #####################################
Line 2: 32/63 ( 50.8%)  ################################
Line 3: 35/63 ( 55.6%)  ###################################
Line 4: 34/63 ( 54.0%)  ##################################
Line 5: 35/63 ( 55.6%)  ###################################
Line 6: 38/63 ( 60.3%)  ######################################

--- Line co-change frequency ---
How often pairs of lines change together in the same transition
      L1   L2   L3   L4   L5   L6
L1    .    18   20   19   19   30
L2    .    .    20   17   23   20
L3    .    .    .    25   16   21
L4    .    .    .    .    20   20
L5    .    .    .    .    .    20
L6    .    .    .    .    .    .
```

Complement distance analysis

```
---
Complement distance analysis
Every hexagram has an 'opposite' (complement) formed by toggling all 6 lines
(solid becomes broken, broken becomes solid). For example, The Creative (all
solid) and The Receptive (all broken) are complements. This section finds where
each hexagram's complement sits in the King Wen sequence and how far apart they
are. If complements sit nearer each other than a null model predicts, opposition
is an organizing feature of the sequence's structure; if farther, complements
may serve as bookends for larger structural sections.
---
01 ䷀ <-> 02 ䷁  distance:  1  The Creative <-> The Receptive
02 ䷁ <-> 01 ䷀  distance:  1  The Receptive <-> The Creative
03 ䷂ <-> 50 ䷱  distance: 47  Difficulty at the Beginning <-> The Cauldron
04 ䷃ <-> 49 ䷰  distance: 45  Youthful Folly <-> Revolution
05 ䷄ <-> 35 ䷢  distance: 30  Waiting <-> Progress
06 ䷅ <-> 36 ䷣  distance: 30  Conflict <-> Darkening of the Light
07 ䷆ <-> 13 ䷌  distance:  6  The Army <-> Fellowship
08 ䷇ <-> 14 ䷍  distance:  6  Holding Together <-> Great Possession
09 ䷈ <-> 16 ䷏  distance:  7  Small Taming <-> Enthusiasm
10 ䷉ <-> 15 ䷎  distance:  5  Treading <-> Modesty
11 ䷊ <-> 12 ䷋  distance:  1  Peace <-> Standstill
12 ䷋ <-> 11 ䷊  distance:  1  Standstill <-> Peace
13 ䷌ <-> 07 ䷆  distance:  6  Fellowship <-> The Army
14 ䷍ <-> 08 ䷇  distance:  6  Great Possession <-> Holding Together
15 ䷎ <-> 10 ䷉  distance:  5  Modesty <-> Treading
16 ䷏ <-> 09 ䷈  distance:  7  Enthusiasm <-> Small Taming
17 ䷐ <-> 18 ䷑  distance:  1  Following <-> Work on the Decayed
18 ䷑ <-> 17 ䷐  distance:  1  Work on the Decayed <-> Following
19 ䷒ <-> 33 ䷠  distance: 14  Approach <-> Retreat
20 ䷓ <-> 34 ䷡  distance: 14  Contemplation <-> Great Power
21 ䷔ <-> 48 ䷯  distance: 27  Biting Through <-> The Well
```

```
22 ䷕ <-> 47 ䷮  distance: 25  Grace <-> Oppression
23 ䷖ <-> 43 ䷪  distance: 20  Splitting Apart <-> Breakthrough
24 ䷗ <-> 44 ䷫  distance: 20  Return <-> Coming to Meet
25 ䷘ <-> 46 ䷭  distance: 21  Innocence <-> Pushing Upward
26 ䷙ <-> 45 ䷬  distance: 19  Great Taming <-> Gathering Together
27 ䷚ <-> 28 ䷛  distance:  1  Nourishment <-> Great Preponderance
28 ䷛ <-> 27 ䷚  distance:  1  Great Preponderance <-> Nourishment
29 ䷜ <-> 30 ䷝  distance:  1  The Abysmal <-> The Clinging
30 ䷝ <-> 29 ䷜  distance:  1  The Clinging <-> The Abysmal
31 ䷞ <-> 41 ䷨  distance: 10  Influence <-> Decrease
32 ䷟ <-> 42 ䷩  distance: 10  Duration <-> Increase
33 ䷠ <-> 19 ䷒  distance: 14  Retreat <-> Approach
34 ䷡ <-> 20 ䷓  distance: 14  Great Power <-> Contemplation
35 ䷢ <-> 05 ䷄  distance: 30  Progress <-> Waiting
36 ䷣ <-> 06 ䷅  distance: 30  Darkening of the Light <-> Conflict
37 ䷤ <-> 40 ䷧  distance:  3  The Family <-> Deliverance
38 ䷥ <-> 39 ䷦  distance:  1  Opposition <-> Obstruction
39 ䷦ <-> 38 ䷥  distance:  1  Obstruction <-> Opposition
40 ䷧ <-> 37 ䷤  distance:  3  Deliverance <-> The Family
41 ䷨ <-> 31 ䷞  distance: 10  Decrease <-> Influence
42 ䷩ <-> 32 ䷟  distance: 10  Increase <-> Duration
43 ䷪ <-> 23 ䷖  distance: 20  Breakthrough <-> Splitting Apart
44 ䷫ <-> 24 ䷗  distance: 20  Coming to Meet <-> Return
45 ䷬ <-> 26 ䷙  distance: 19  Gathering Together <-> Great Taming
46 ䷭ <-> 25 ䷘  distance: 21  Pushing Upward <-> Innocence
47 ䷮ <-> 22 ䷕  distance: 25  Oppression <-> Grace
48 ䷯ <-> 21 ䷔  distance: 27  The Well <-> Biting Through
49 ䷰ <-> 04 ䷃  distance: 45  Revolution <-> Youthful Folly
50 ䷱ <-> 03 ䷂  distance: 47  The Cauldron <-> Difficulty at the Beginning
51 ䷲ <-> 57 ䷸  distance:  6  The Arousing <-> The Gentle
52 ䷳ <-> 58 ䷹  distance:  6  Keeping Still <-> The Joyous
53 ䷴ <-> 54 ䷵  distance:  1  Development <-> The Marrying Maiden
54 ䷵ <-> 53 ䷴  distance:  1  The Marrying Maiden <-> Development
55 ䷶ <-> 59 ䷺  distance:  4  Abundance <-> Dispersion
56 ䷷ <-> 60 ䷻  distance:  4  The Wanderer <-> Limitation
57 ䷸ <-> 51 ䷲  distance:  6  The Gentle <-> The Arousing
58 ䷹ <-> 52 ䷳  distance:  6  The Joyous <-> Keeping Still
59 ䷺ <-> 55 ䷶  distance:  4  Dispersion <-> Abundance
60 ䷻ <-> 56 ䷷  distance:  4  Limitation <-> The Wanderer
61 ䷼ <-> 62 ䷽  distance:  1  Inner Truth <-> Small Preponderance
62 ䷽ <-> 61 ䷼  distance:  1  Small Preponderance <-> Inner Truth
63 ䷾ <-> 64 ䷿  distance:  1  After Completion <-> Before Completion
64 ䷿ <-> 63 ䷾  distance:  1  Before Completion <-> After Completion

--- Complement distance statistics ---
Mean distance:   12.1
Median distance: 6
Min distance:    1 (adjacent = pair is an inverse pair)
Max distance:    47

--- Complement distance null model ---
Random mean complement distance (over 10000 shuffles): 21.7
King Wen mean complement distance: 12.1
King Wen percentile vs random: 0.0%
Complements are significantly closer together than chance would predict:
the ordering keeps opposites unusually near one another.
```

Palindrome analysis of the difference wave

```
---
Palindrome analysis of the difference wave
A palindrome reads the same forwards and backwards (like 2,4,6,4,2). Palindromic
runs in the difference wave are mirror symmetry — sections where the pattern of
change rises and falls in a balanced way. Longer palindromes are less likely to
occur by chance, so counts and lengths are measured against null models below.
---
Found 27 palindromic subsequences (showing longest 20)

Length  7 at positions 11-17: [6,2,2,4,2,2,6] █▂▂▅▂▂█
Length  7 at positions 19-25: [4,3,2,2,2,3,4] ▅▃▂▂▂▃▅
Length  7 at positions 39-45: [4,3,2,4,2,3,4] ▅▃▂▅▂▃▅
```

```
Length  5 at positions 12-16: [2,2,4,2,2] ▂▂▅▂▂
Length  5 at positions 20-24: [3,2,2,2,3] ▃▂▂▂▃
Length  5 at positions 40-44: [3,2,4,2,3] ▃▂▅▂▃
Length  5 at positions 43-47: [2,3,4,3,2] ▂▃▅▃▂
Length  5 at positions 45-49: [4,3,2,3,4] ▅▃▂▃▅
Length  5 at positions 55-59: [2,3,4,3,2] ▂▃▅▃▂
Length  3 at positions 03-05: [4,4,4] ▅▅▅
Length  3 at positions 07-09: [2,4,2] ▂▅▂
Length  3 at positions 08-10: [4,2,4] ▅▂▅
Length  3 at positions 13-15: [2,4,2] ▂▅▂
Length  3 at positions 18-20: [3,4,3] ▃▅▃
Length  3 at positions 21-23: [2,2,2] ▂▂▂
Length  3 at positions 26-28: [2,6,2] ▂█▂
Length  3 at positions 27-29: [6,2,6] █▂█
Length  3 at positions 30-32: [3,2,3] ▃▂▃
Length  3 at positions 33-35: [4,4,4] ▅▅▅
Length  3 at positions 35-37: [4,2,4] ▅▂▅

--- Palindrome count by length ---
Length  7: 3 found
Length  5: 6 found
Length  3: 18 found

--- Palindrome null model ---
How many palindromes of length >= 3 do random permutations produce?
King Wen palindromes: 27, longest: 7
Random mean palindromes: 22.3, mean longest: 6.9
King Wen palindrome count percentile: 79.1%
King Wen longest palindrome percentile: 42.7%

--- Pair-constrained palindrome null model ---
The unconstrained comparison shuffles all 64 hexagrams freely. A fairer
comparison preserves the pair structure: each pair of hexagrams stays
adjacent, only the pair order and within-pair orientation are randomized.
King Wen palindromes: 27, longest: 7
Pair-constrained mean palindromes: 27.6, mean longest: 8.4
King Wen palindrome count percentile (pair-constrained): 49.4%
King Wen longest palindrome percentile (pair-constrained): 13.7%
```

Upper Canon (1-30) vs. Lower Canon (31-64) comparison

```
---
Upper Canon (1-30) vs. Lower Canon (31-64) comparison
The I Ching is traditionally divided into two books: the Upper Canon (hexagrams
1-30) and Lower Canon (hexagrams 31-64). This division is ancient and may reflect
different authorship, time periods, or thematic groupings. This section compares
the mathematical properties of each half: do they have similar difference wave
patterns? Similar yin/yang balance? Does the 'no 5-line transition' property hold
in each half independently, or only in the whole?
---
Metric                         Upper (1-30)  Lower (31-64)
------------------------------ ------------ --------------
Hexagrams                                30            34
Transitions                              29            33
Mean line changes                      3.38          3.33
Yang lines (total)                       86           106
Yin lines (total)                        94            98

Difference distribution:       Upper (1-30)  Lower (31-64)
0 line changes                            0             0
1 line changes                            0             2
2 line changes                           12             8
3 line changes                            4             8
4 line changes                            8            11
5 line changes                            0             0
6 line changes                            5             4

Upper Canon wave: █▂▅▅▅▃▂▅▂▅█▂▂▅▂▂█▃▅▃▂▂▂▃▅▂█▂█
Lower Canon wave: ▂▃▅▅▅▂▅█▅▃▂▅▂▃▅▃▂▃▅▅▅▁█▂▂▃▅▃▂▁█▃█

5-line transitions in Upper Canon: 0
```

```
5-line transitions in Lower Canon: 0

--- Canon split null model ---
Permutation test: is the King Wen split at position 30 special, or would
any random split of the 64-hexagram sequence show a similar gap in mean
line-change differences between the two halves?
King Wen |upper_mean - lower_mean|: 0.0460
Random permutations with gap >= King Wen: 8764/10000
King Wen gap percentile: 12.4%
The King Wen canon split does not produce a statistically significant
mean-difference gap. The observed difference between upper and lower
halves is consistent with what random shuffles produce.
```

Autocorrelation of the difference wave

```
---
Autocorrelation of the difference wave
Autocorrelation measures whether the wave is correlated with a shifted copy of
itself. A peak at lag N means the pattern tends to repeat every N steps. A value
near +1 means strong repetition, near -1 means strong alternation, and near 0
means no relationship. If the King Wen ordering contains hidden periodic
structure, it will show up as peaks in this analysis.
Caveat: with only N=63 data points, the 95% noise threshold is +/-0.25.
Values within that band are indistinguishable from random noise.
---
Mean difference: 3.349
Variance: 1.910
95% noise threshold: +/-0.247 (values inside this are not significant)

Lag    Autocorrelation  Sig?  Visualization
---    ---------------  ----  -------------
0               1.0000        [            :  |  :          #]
1              -0.2469        [            #  |  :           ]
2              -0.0098        [            :  |  :           ]
3              -0.0491        [            :  |  :           ]
4              -0.0447        [            :  |  :           ]
5              -0.0457        [            :  |  :           ]
6              -0.0218        [            :  |  :           ]
7              -0.0644        [            :  |  :           ]
8               0.1204        [            :  |# :           ]
9              -0.1275        [            : #|  :           ]
10              0.1375        [            :  | #:           ]
11             -0.0290        [            :  |  :           ]
12             -0.0221        [            :  |  :           ]
13              0.0850        [            :  |# :           ]
14             -0.1596        [            :# |  :           ]
15             -0.0110        [            :  |  :           ]
16              0.1015        [            :  |# :           ]
17             -0.1514        [            :# |  :           ]
18              0.0055        [            :  |  :           ]
19              0.0851        [            :  |# :           ]
20             -0.1403        [            :# |  :           ]
21              0.0274        [            :  |  :           ]
22             -0.1315        [            : #|  :           ]
23             -0.0137        [            :  |  :           ]
24              0.2263        [            :  |  #           ]
25             -0.1595        [            :# |  :           ]
26              0.1112        [            :  |# :           ]
27             -0.1109        [            : #|  :           ]
28              0.1649        [            :  | #:           ]
29              0.0093        [            :  |  :           ]
30             -0.0969        [            : #|  :           ]
31             -0.0480        [            :  |  :           ]

Significant lags (outside noise band): 0/31
(Expected false positives at 95% threshold with 31 lags: ~1.6)
No significant autocorrelation detected — consistent with no hidden periodicity,
but note that N=63 provides limited statistical power to detect weak periodicity.
```

Spectral analysis (Discrete Fourier Transform)

```
---
Spectral analysis (Discrete Fourier Transform)
The DFT decomposes the difference wave into frequency components, like splitting
white light into a rainbow. Each frequency represents a periodic pattern in the
wave. A strong peak at frequency F means the wave has a repeating cycle every
63/F transitions. If the King Wen ordering has hidden periodic structure, the
spectrum will show clear peaks rather than flat noise.
Caveat: with only 63 samples, each of the 31 frequency bins has wide
uncertainty. A flat spectrum may indicate no periodicity, or insufficient data.
---
Number of samples: 63
Mean (DC component): 3.349
White noise floor: 0.1741 (magnitudes near this are not significant)

Freq   Period  Magnitude  Sig?  Spectrum
----   ------  ---------  ----  --------
   1     63.0     0.0224        ##
   2     31.5     0.1823        #################
   3     21.0     0.0145        #
   4     15.8     0.1359        ############
   5     12.6     0.2021        ##################
   6     10.5     0.1157        ##########
   7      9.0     0.2399        ######################
   8      7.9     0.1092        ##########
   9      7.0     0.0553        #####
  10      6.3     0.1630        ###############
  11      5.7     0.2371        ######################
  12      5.2     0.0473        ####
  13      4.8     0.2326        #####################
  14      4.5     0.0331        ###
  15      4.2     0.1145        ##########
  16      3.9     0.2204        ####################
  17      3.7     0.1451        #############
  18      3.5     0.1975        ##################
  19      3.3     0.2197        ####################
  20      3.1     0.2166        ####################
  21      3.0     0.1611        ###############
  22      2.9     0.0222        ##
  23      2.7     0.1210        ###########
  24      2.6     0.4266     *  ########################################
  25      2.5     0.0306        ##
  26      2.4     0.2297        #####################
  27      2.3     0.0795        #######
  28      2.2     0.0639        #####
  29      2.2     0.2114        ###################
  30      2.1     0.2129        ###################
  31      2.0     0.2033        ###################

Frequencies above 2x noise floor: 1/31
(The 2x threshold is ad hoc. A proper test would use Fisher's g-statistic or
Bonferroni correction across frequency bins. With only 63 samples, even real
periodicity may not rise above the noise floor.)
```

Markov chain analysis

```
---
Markov chain analysis
If we treat each difference value as a 'state', we can ask: does the current
difference predict the next one? In a random sequence, knowing that the last
transition changed 2 lines tells you nothing about the next transition. But if
certain patterns like '6 is always followed by 2' emerge, it reveals hidden
rules in the King Wen ordering. The matrix shows the probability of each
transition between difference values.
---
Transition probability matrix (row = from, column = to)
        -> 1    -> 2    -> 3    -> 4    -> 6
  1     0.00    0.00    0.00    0.00    1.00    (n=2)
```

```
  2     0.05    0.25    0.25    0.30    0.15    (n=20)
  3     0.00    0.46    0.00    0.46    0.08    (n=13)
  4     0.05    0.26    0.26    0.32    0.11    (n=19)
  6     0.00    0.50    0.38    0.12    0.00    (n=8)

--- Notable transition patterns ---
(Caveat: rows with small n are unreliable — patterns based on <5 observations
are anecdotes, not statistically meaningful.)
  3 -> 2: 46% (6/13 times)
  3 -> 4: 46% (6/13 times)

--- Permutation test ---
Is the transition matrix more concentrated than random orderings produce?
King Wen matrix concentration: 2.3303
Random orderings equally or more concentrated: 2087/5000 (41.7%)
```

Gray code comparison

```
---
Gray code comparison
A Gray code is an ordering of binary values where each consecutive pair differs
by exactly 1 bit. For 6-bit hexagrams, a Gray code path would change exactly 1
line at every step — the smoothest possible sequence. The King Wen sequence does
NOT follow a Gray code (most transitions change 2-6 lines). This comparison
shows how much 'rougher' King Wen is than the theoretical minimum, suggesting
the ordering prioritizes something other than smooth transitions.
---
Metric                               Gray Code   King Wen
----------------------------------- ---------- ----------
Total path length                           63        211
Mean change per transition                1.00       3.35
Max single transition                        1          6
Min single transition                        1          1
King Wen / Gray Code ratio                             3.35x

Difference distribution:             Gray Code   King Wen
  1 line changes                             63          2
  2 line changes                              0         20
  3 line changes                              0         13
  4 line changes                              0         19
  6 line changes                              0          9

Gray code wave: ▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁▁
King Wen wave:  █▂▅▅▅▃▂▅▂▅█▂▂▅▂▂█▃▅▃▂▂▂▃▅▂█▂█▃▂▃▅▅▅▂▅█▅▃▂▅▂▃▅▃▂▃▅▅▅▁█▂▂▃▅▃▂▁█▃█

--- Methodological note ---
This comparison is descriptive. The Gray code ratio (King Wen path / Gray
code path) is not a statistical test. For significance testing of King Wen's
path length against appropriate null models, see --path.
```

Symmetry group analysis (XOR algebra)

```
---
Symmetry group analysis (XOR algebra)
The 64 hexagrams form a mathematical group under XOR: combining any two hexagrams
by toggling their differing lines always produces another valid hexagram. This is
like mixing colors — every combination is itself a color. Under this algebra,
certain sets of hexagrams form 'subgroups' (closed sets where combining any two
members stays in the set). This section checks whether the King Wen pairs, inverse
pairs, and the canonical divisions correspond to meaningful algebraic subgroups.
---
XOR identity element: 0b000000 (hexagram 2, The Receptive)

--- Fixed points under complement (XOR 0b111111) ---
Hexagrams that are their own complement do not exist (since XOR with
0b111111 always flips all bits, no 6-bit value equals its own complement).

--- King Wen pair XOR products ---
```

```
For each pair, XOR the two hexagrams to see what 'difference element' they produce.
If many pairs produce the same XOR product, the pairing has algebraic regularity.

XOR Product      Hexagram  Pair count  Pairs
  001100   62 ䷽       4      9-10, 15-16, 21-22, 47-48
  010010   29 ䷜       4      7-8, 13-14, 31-32, 41-42
  011110   28 ䷛       4      25-26, 37-38, 39-40, 45-46
  100001   27 ䷚       4      23-24, 43-44, 55-56, 59-60
  101101   30 ䷝       4      5-6, 35-36, 51-52, 57-58
  110011   61 ䷼       4      3-4, 19-20, 33-34, 49-50
  111111   01 ䷀       8      1-2, 11-12, 17-18, 27-28, 29-30, 53-54, 61-62, 63-64

Unique XOR products: 7 (out of 32 pairs)
If this number is small, the pairs share algebraic structure.

--- Inverse pair XOR closure test ---
Inverse pair hexagrams (16 total): 01, 02, 11, 12, 17, 18, 27, 28, 29, 30, 53, 54, 61, 62, 63, 64
Closed under XOR: Yes — forms a subgroup
```

Alternative sequence comparison

```
---
Alternative sequence comparison
The King Wen ordering is not the only way to arrange 64 hexagrams. The Fu Xi
(binary) sequence orders them by numerical value (0-63), which is mathematically
natural but has no traditional significance. The Mawangdui sequence was found on
silk manuscripts in a 168 BCE tomb and may represent an independent tradition.
Comparing the same analyses across orderings reveals what is unique to King Wen.
---
Metric                                King Wen   Fu Xi (binary)        Mawangdui
------------------------- ---------------- ---------------- ----------------
Total path length                          211              120              141
Mean change                               3.35             1.90             2.24
1-line transitions                           2               32               21
2-line transitions                          20               16               10
3-line transitions                          13                8               29
4-line transitions                          19                4                2
5-line transitions                           0                2                1
6-line transitions                           9                1                0

Waves:
  King Wen
  Fu Xi (binary)
  Mawangdui
```

```
--- What makes King Wen unique ---
Zero 5-line transitions: King Wen=0, Fu Xi=2, Mawangdui=1
Zero 0-line transitions: King Wen=0, Fu Xi=0, Mawangdui=0
```

Windowed entropy analysis

```
---
Windowed entropy analysis
Instead of one entropy value for the whole wave, we slide a window across it
and compute entropy locally. This reveals WHERE in the sequence structure
concentrates (low entropy = predictable) vs. dissolves (high entropy = varied).
Dips in the curve mark regions with repetitive transition patterns.
---
Window size: 15
Center  Entropy    Visualization
------  -------    -------------
     8   1.7056   ########################
     9   1.5628   ######################
    10   1.7056   ########################
    11   1.8323   ##########################
    12   1.8323   ##########################
    13   1.8892   ##########################
    14   1.7968   #########################
```

```
    15   1.7968   #########################
    16   1.7232   ########################
    17   1.8295   ##########################
    18   1.8295   ##########################
    19   1.6729   #######################
    20   1.8295   ##########################
    21   1.8295   ##########################
    22   1.8295   ##########################
    23   1.8892   ##########################
    24   1.8892   ##########################
    25   1.8323   ##########################
    26   1.8892   ##########################
    27   1.8892   ##########################
    28   1.8892   ##########################
    29   1.8892   ##########################
    30   1.9086   ###########################
    31   1.9656   ############################
    32   1.8892   ##########################
    33   1.9656   ############################
    34   1.9656   ############################
    35   1.8892   ##########################
    36   1.8892   ##########################
    37   1.8062   #########################
    38   1.7465   ########################
    39   1.7465   ########################
    40   1.7465   ########################
    41   1.8062   #########################
    42   1.8062   #########################
    43   1.8062   #########################
    44   1.7465   ########################
    45   2.0226   ############################
    46   2.0226   ############################
    47   2.0662   #############################
    48   2.0419   #############################
    49   2.0662   #############################
    50   2.0662   #############################
    51   2.0419   #############################
    52   2.0662   #############################
    53   2.1736   ##############################
    54   2.2566   ################################
    55   2.2566   ################################
    56   2.2892   ################################

Mean windowed entropy: 1.8956
Min entropy: 1.5628 at position 9 (most structured)
Max entropy: 2.2892 at position 56 (most varied)

Entropy spark: ▃▃▃▃▃▄▃▃▃▃▃▃▃▃▃▄▄▃▄▄▄▄▄▄▄▄▄▄▄▃▃▃▃▃▃▃▃▄▄▄▄▄▄▄▄▄▄▄▄

--- Methodological note ---
Windowed entropy is an exploratory visualization, not a statistical test.
Apparent patterns (dips and peaks) are expected from random variation in a
short sequence. Without a null model, no significance can be assigned to
specific regions. Interpret as descriptive, not inferential.
```

Mutual information: upper vs. lower trigram transitions

```
---
Mutual information: upper vs. lower trigram transitions
When two consecutive hexagrams differ, do the upper and lower trigrams change
independently, or is knowing that the upper trigram changed informative about
whether the lower trigram also changed? Mutual information quantifies this:
  0 bits = completely independent (knowing one tells you nothing about the other)
  higher = correlated (they tend to change together or stay together)
---
Joint distribution of trigram changes:
                       Lower unchanged  Lower changed
  Upper unchanged          0 (  0.0%)         4 (  6.3%)
  Upper changed            5 (  7.9%)        54 ( 85.7%)

Mutual information: 0.0078 bits
```

```
Normalized MI: 0.0230 (0=independent, 1=perfectly correlated)

Mean MI of random permutations: 0.0202 bits
King Wen percentile: 7.3%

--- Full 8-state trigram mutual information ---
Using actual trigram identities (8 possible values each) rather than
binary changed/unchanged, which preserves more information.
MI(upper, lower) across all 64 hexagrams: 0.000000 bits
(Expected: 0.0 — since all 64 combinations appear exactly once,
the trigrams are perfectly independent in the static set.)
Note: with all 64 upper/lower trigram combinations present exactly once,
independence is expected by construction (complete Latin square).
```

Yin-yang balance wave

```
---
Yin-yang balance wave
Each hexagram has 6 lines that are either yang (solid, 1) or yin (broken, 0).
Note: since the King Wen sequence contains all 64 possible hexagrams exactly
once, the total yin-yang count is necessarily balanced (192 each). This is a
mathematical tautology, not a property of the ordering. However, the LOCAL
balance as you move through the sequence IS a property of the ordering — it
shows where yang-dominant or yin-dominant hexagrams cluster together.
---
Total yang lines: 192 / 384 (50.0%)
Total yin lines:  192 / 384 (50.0%)
Mean yang per hexagram: 3.00

Running yang average (window=8):
 Pos  Yang  RunAvg     Visualization
   1     6    6.00   [                    |                   #]
   2     0    3.00   [                    #                    ]
   3     2    2.67   [                 #  |                    ]
   4     2    2.50   [                #   |                    ]
   5     4    2.80   [                  # |                    ]
   6     4    3.00   [                    #                    ]
   7     1    2.71   [                  # |                    ]
   8     1    2.50   [                #   |                    ]
   9     5    2.38   [               #    |                    ]
  10     5    3.00   [                    #                    ]
  11     3    3.12   [                    #                    ]
  12     3    3.25   [                    |#                   ]
  13     5    3.38   [                    | #                  ]
  14     5    3.50   [                    |  #                 ]
  15     1    3.50   [                    |  #                 ]
  16     1    3.50   [                    |  #                 ]
  17     3    3.25   [                    |#                   ]
  18     3    3.00   [                    #                    ]
  19     2    2.88   [                   #|                    ]
  20     2    2.75   [                  # |                    ]
  21     3    2.50   [                #   |                    ]
  22     3    2.25   [               #    |                    ]
  23     1    2.25   [               #    |                    ]
  24     1    2.25   [               #    |                    ]
  25     4    2.38   [               #    |                    ]
  26     4    2.50   [                #   |                    ]
  27     2    2.50   [                #   |                    ]
  28     4    2.75   [                  # |                    ]
  29     2    2.62   [                 #  |                    ]
  30     4    2.75   [                  # |                    ]
  31     3    3.00   [                    #                    ]
  32     3    3.25   [                    |#                   ]
  33     4    3.25   [                    |#                   ]
  34     4    3.25   [                    |#                   ]
  35     2    3.25   [                    |#                   ]
  36     2    3.00   [                    #                    ]
  37     4    3.25   [                    |#                   ]
  38     4    3.25   [                    |#                   ]
  39     2    3.12   [                    #                    ]
  40     2    3.00   [                    #                    ]
```

```
  41     3    2.88   [                   #|                    ]
  42     3    2.75   [                  # |                    ]
  43     5    3.12   [                    #                    ]
  44     5    3.50   [                    |  #                 ]
  45     2    3.25   [                    |#                   ]
  46     2    3.00   [                    #                    ]
  47     3    3.12   [                    #                    ]
  48     3    3.25   [                    |#                   ]
  49     4    3.38   [                    | #                  ]
  50     4    3.50   [                    |  #                 ]
  51     2    3.12   [                    #                    ]
  52     2    2.75   [                  # |                    ]
  53     3    2.88   [                   #|                    ]
  54     3    3.00   [                    #                    ]
  55     3    3.00   [                    #                    ]
  56     3    3.00   [                    #                    ]
  57     4    3.00   [                    #                    ]
  58     4    3.00   [                    #                    ]
  59     3    3.12   [                    #                    ]
  60     3    3.25   [                    |#                   ]
  61     4    3.38   [                    | #                  ]
  62     2    3.25   [                    |#                   ]
  63     3    3.25   [                    |#                   ]
  64     3    3.25   [                    |#                   ]

Yang count spark: █ ▂▂▅▅▁▁▆▆▃▃▆▆▁▁▃▃▂▂▃▃▁▁▅▅▂▅▂▅▃▃▅▅▂▂▅▅▂▂▃▃▆▆▂▂▃▃▅▅▂▂▃▃▃▃▅▅▃▃▅▂▃▃

--- Yang/Yin dominance runs ---
  Positions 02-04: 3 consecutive yin-dominant hexagrams
```

Odd-vs-even transition parity (McKenna 25/75)

```
---
Odd-vs-even transition parity (McKenna's 25/75 observation)
---
Each transition d(s_i, s_{i+1}) is between 1 and 6 line changes.
Theorem 1 says within-pair distances are always even (in {2,4,6}),
so all 32 within-pair transitions are forced even by C1. The 31
between-pair transitions can be any non-5 value.

McKenna (The Invisible Landscape, 1975) noted that ~25% of King Wen's
transitions are odd-distance and ~75% are even-distance. We report both
the LINEAR reading (the standard 63-transition difference wave) and the
CIRCULAR reading (64 transitions, including the wrap-around s_63 -> s_0).

Last hexagram (position 64): #42 (101010)
First hexagram (position 1): #63 (111111)
Wrap-around distance: hamming(#42, #63) = 3 (odd)

Mode                       Total    Odd       %Odd   Even      %Even
LINEAR (63 trans)             63     15   23.8095%     48   76.1905%
CIRCULAR (64 trans)           64     16   25.0000%     48   75.0000%

CIRCULAR parity is EXACTLY 25.0000% / 75.0000% — McKenna's 25/75 holds exactly
when the wrap-around transition is included. Without wrap-around, the linear
split is 23.8095% / 76.1905% — approximately but not exactly 25/75.

--- Structural note ---
All 32 within-pair transitions are forced even by C1 + Theorem 1.
So the parity split is determined entirely by the 31 (or 32 with wrap)
between-pair transitions:
  Between-pair transitions: 15 odd, 16 even
  Within-pair transitions: 0 odd (forced), 32 even (forced)
  Plus wrap-around: 1 odd

--- 'C8' candidate: is the wrap-around parity a structural constraint? ---
The spec's C1-C7 do not constrain the wrap-around s_63 -> s_0. So if
McKenna's exact 25/75 reflects a real constraint on the sequence rather
than a numerical coincidence, it would imply a new constraint:
  C8 (weak):   hamming(s_63, s_0) is odd  (admits {1, 3, 5}; C2 may extend to 64th)
  C8 (strict): hamming(s_63, s_0) = 3  (King Wen's exact value)
```

```
Restrictiveness: the prior for wrap-around distance among {1,2,3,4,6} (5 forbidden
by C2 if extended, 0 impossible for a permutation) is uniform → P(odd) ≈ 40%, so
the weak form eliminates ~60% of completions; the strict form eliminates ~80%.

Whether C8 is a real constraint or a numerical coincidence is empirical: a Monte
Carlo sample of sequences satisfying C1-C7 would tell us how non-uniform the
wrap-around distribution actually is among King Wen-eligible orderings. Today
this is NOT in the formal spec — it remains an observation, not a constraint.
```

Hexagram neighborhoods

```
---
Hexagram neighborhoods
For each hexagram, which others are 'nearby' (differ by only 1 or 2 lines)?
Dense neighborhoods mean a hexagram has many close relatives; sparse ones are
more isolated. This also shows whether King Wen places neighbors near each
other in the sequence or scatters them.
---
01 ䷀ The Creative                 d=1: [09, 10, 13, 14, 43, 44] (6 neighbors, mean seq dist: 21.2)
02 ䷁ The Receptive                d=1: [07, 08, 15, 16, 23, 24] (6 neighbors, mean seq dist: 13.5)
03 ䷂ Difficulty at the Beginning  d=1: [08, 17, 24, 42, 60, 63] (6 neighbors, mean seq dist: 32.7)
04 ䷃ Youthful Folly               d=1: [07, 18, 23, 41, 59, 64] (6 neighbors, mean seq dist: 31.3)
05 ䷄ Waiting                      d=1: [09, 11, 43, 48, 60, 63] (6 neighbors, mean seq dist: 34.0)
06 ䷅ Conflict                     d=1: [10, 12, 44, 47, 59, 64] (6 neighbors, mean seq dist: 33.3)
07 ䷆ The Army                     d=1: [02, 04, 19, 29, 40, 46] (6 neighbors, mean seq dist: 19.0)
08 ䷇ Holding Together             d=1: [02, 03, 20, 29, 39, 45] (6 neighbors, mean seq dist: 18.7)
09 ䷈ Small Taming                 d=1: [01, 05, 26, 37, 57, 61] (6 neighbors, mean seq dist: 26.2)
10 ䷉ Treading                     d=1: [01, 06, 25, 38, 58, 61] (6 neighbors, mean seq dist: 25.8)
11 ䷊ Peace                        d=1: [05, 19, 26, 34, 36, 46] (6 neighbors, mean seq dist: 18.7)
12 ䷋ Standstill                   d=1: [06, 20, 25, 33, 35, 45] (6 neighbors, mean seq dist: 17.3)
13 ䷌ Fellowship                   d=1: [01, 25, 30, 33, 37, 49] (6 neighbors, mean seq dist: 20.2)
14 ䷍ Great Possession             d=1: [01, 26, 30, 34, 38, 50] (6 neighbors, mean seq dist: 20.2)
15 ䷎ Modesty                      d=1: [02, 36, 39, 46, 52, 62] (6 neighbors, mean seq dist: 28.8)
16 ䷏ Enthusiasm                   d=1: [02, 35, 40, 45, 51, 62] (6 neighbors, mean seq dist: 27.8)
17 ䷐ Following                    d=1: [03, 25, 45, 49, 51, 58] (6 neighbors, mean seq dist: 26.2)
18 ䷑ Work on the Decayed          d=1: [04, 26, 46, 50, 52, 57] (6 neighbors, mean seq dist: 25.8)
19 ䷒ Approach                     d=1: [07, 11, 24, 41, 54, 60] (6 neighbors, mean seq dist: 20.5)
20 ䷓ Contemplation                d=1: [08, 12, 23, 42, 53, 59] (6 neighbors, mean seq dist: 19.5)
21 ䷔ Biting Through               d=1: [25, 27, 30, 35, 38, 51] (6 neighbors, mean seq dist: 13.3)
22 ䷕ Grace                        d=1: [26, 27, 30, 36, 37, 52] (6 neighbors, mean seq dist: 12.7)
23 ䷖ Splitting Apart              d=1: [02, 04, 20, 27, 35, 52] (6 neighbors, mean seq dist: 14.7)
24 ䷗ Return                       d=1: [02, 03, 19, 27, 36, 51] (6 neighbors, mean seq dist: 15.0)
25 ䷘ Innocence                    d=1: [10, 12, 13, 17, 21, 42] (6 neighbors, mean seq dist: 11.5)
26 ䷙ Great Taming                 d=1: [09, 11, 14, 18, 22, 41] (6 neighbors, mean seq dist: 11.8)
27 ䷚ Nourishment                  d=1: [21, 22, 23, 24, 41, 42] (6 neighbors, mean seq dist: 7.8)
28 ䷛ Great Preponderance          d=1: [31, 32, 43, 44, 47, 48] (6 neighbors, mean seq dist: 12.8)
29 ䷜ The Abysmal                  d=1: [07, 08, 47, 48, 59, 60] (6 neighbors, mean seq dist: 23.5)
30 ䷝ The Clinging                 d=1: [13, 14, 21, 22, 55, 56] (6 neighbors, mean seq dist: 16.8)
31 ䷞ Influence                    d=1: [28, 33, 39, 45, 49, 62] (6 neighbors, mean seq dist: 12.7)
32 ䷟ Duration                     d=1: [28, 34, 40, 46, 50, 62] (6 neighbors, mean seq dist: 12.7)
33 ䷠ Retreat                      d=1: [12, 13, 31, 44, 53, 56] (6 neighbors, mean seq dist: 16.2)
34 ䷡ Great Power                  d=1: [11, 14, 32, 43, 54, 55] (6 neighbors, mean seq dist: 15.8)
35 ䷢ Progress                     d=1: [12, 16, 21, 23, 56, 64] (6 neighbors, mean seq dist: 19.7)
36 ䷣ Darkening of the Light       d=1: [11, 15, 22, 24, 55, 63] (6 neighbors, mean seq dist: 19.7)
37 ䷤ The Family                   d=1: [09, 13, 22, 42, 53, 63] (6 neighbors, mean seq dist: 19.0)
38 ䷥ Opposition                   d=1: [10, 14, 21, 41, 54, 64] (6 neighbors, mean seq dist: 19.0)
39 ䷦ Obstruction                  d=1: [08, 15, 31, 48, 53, 63] (6 neighbors, mean seq dist: 18.3)
40 ䷧ Deliverance                  d=1: [07, 16, 32, 47, 54, 64] (6 neighbors, mean seq dist: 18.3)
41 ䷨ Decrease                     d=1: [04, 19, 26, 27, 38, 61] (6 neighbors, mean seq dist: 18.5)
42 ䷩ Increase                     d=1: [03, 20, 25, 27, 37, 61] (6 neighbors, mean seq dist: 19.5)
43 ䷪ Breakthrough                 d=1: [01, 05, 28, 34, 49, 58] (6 neighbors, mean seq dist: 20.8)
44 ䷫ Coming to Meet               d=1: [01, 06, 28, 33, 50, 57] (6 neighbors, mean seq dist: 21.2)
45 ䷬ Gathering Together           d=1: [08, 12, 16, 17, 31, 47] (6 neighbors, mean seq dist: 23.8)
46 ䷭ Pushing Upward               d=1: [07, 11, 15, 18, 32, 48] (6 neighbors, mean seq dist: 24.8)
47 ䷮ Oppression                   d=1: [06, 28, 29, 40, 45, 58] (6 neighbors, mean seq dist: 16.3)
48 ䷯ The Well                     d=1: [05, 28, 29, 39, 46, 57] (6 neighbors, mean seq dist: 17.0)
49 ䷰ Revolution                   d=1: [13, 17, 31, 43, 55, 63] (6 neighbors, mean seq dist: 18.7)
50 ䷱ The Cauldron                 d=1: [14, 18, 32, 44, 56, 64] (6 neighbors, mean seq dist: 18.7)
51 ䷲ The Arousing                 d=1: [16, 17, 21, 24, 54, 55] (6 neighbors, mean seq dist: 22.2)
52 ䷳ Keeping Still               d=1: [15, 18, 22, 23, 53, 56] (6 neighbors, mean seq dist: 22.5)
```

```
53 ䷴ Development                  d=1: [20, 33, 37, 39, 52, 57] (6 neighbors, mean seq dist: 14.7)
54 ䷵ The Marrying Maiden          d=1: [19, 34, 38, 40, 51, 58] (6 neighbors, mean seq dist: 15.3)
55 ䷶ Abundance                    d=1: [30, 34, 36, 49, 51, 62] (6 neighbors, mean seq dist: 13.7)
56 ䷷ The Wanderer                 d=1: [30, 33, 35, 50, 52, 62] (6 neighbors, mean seq dist: 14.3)
57 ䷸ The Gentle                   d=1: [09, 18, 44, 48, 53, 59] (6 neighbors, mean seq dist: 19.2)
58 ䷹ The Joyous                   d=1: [10, 17, 43, 47, 54, 60] (6 neighbors, mean seq dist: 20.2)
59 ䷺ Dispersion                   d=1: [04, 06, 20, 29, 57, 61] (6 neighbors, mean seq dist: 30.2)
60 ䷻ Limitation                   d=1: [03, 05, 19, 29, 58, 61] (6 neighbors, mean seq dist: 31.2)
61 ䷼ Inner Truth                  d=1: [09, 10, 41, 42, 59, 60] (6 neighbors, mean seq dist: 24.2)
62 ䷽ Small Preponderance          d=1: [15, 16, 31, 32, 55, 56] (6 neighbors, mean seq dist: 27.8)
63 ䷾ After Completion             d=1: [03, 05, 36, 37, 39, 49] (6 neighbors, mean seq dist: 34.8)
64 ䷿ Before Completion            d=1: [04, 06, 35, 38, 40, 50] (6 neighbors, mean seq dist: 35.2)

--- Neighborhood clustering in King Wen sequence ---
Mean sequence distance between Hamming-1 neighbors: 20.6
Mean sequence distance between any two hexagrams:   21.7
Hamming-1 neighbors are closer than average in the sequence (clustered).

--- Neighborhood clustering null model ---
Shuffling binary hexagrams 10,000 times to build a null distribution
of mean sequence distance between Hamming-1 neighbors.
Null distribution: min=18.3, mean=21.7, max=25.2
King Wen observed mean: 20.6
Percentile: 12.3% (proportion of shuffles with mean <= King Wen's)
King Wen's neighborhood clustering is within the range expected by chance.
```

Recurrence plot

```
---
Recurrence plot
A recurrence plot shows where the difference wave repeats itself. Each cell (i,j)
is marked if the difference at position i equals the difference at position j.
Diagonal lines indicate repeated sequences; vertical/horizontal lines indicate
values that persist. Clusters of marks reveal recurring local patterns.
---
             1         2         3         4         5         6
    123456789012345678901234567890123456789012345678901234567890123
  1 +.........#.....#.........#.#........#..............#.......#.#
  2 .+....#.#..##.##....###..#.#..#....#....#.#...#......##...#....
  3 ..+##..#.#...#....#.....#.......###.#.#..#..#...###.....#......
  4 ..#+#..#.#...#....#.....#.......###.#.#..#..#...###.....#......
  5 ..##+..#.#...#....#.....#.......###.#.#..#..#...###.....#......
  6 .....+...........#.#...#.....#.#.......#...#.#.#.......#.#...#.
  7 .#....+.#..##.##....###..#.#..#....#....#.#...#......##...#....
  8 ..###..+.#...#....#.....#.......###.#.#..#..#...###.....#......
  9 .#....#.+..##.##....###..#.#..#....#....#.#...#......##...#....
 10 ..###..#.+...#....#.....#.......###.#.#..#..#...###.....#......
 11 #.........+.....#.........#.#........#..............#.......#.#
 12 .#....#.#..+#.##....###..#.#..#....#....#.#...#......##...#....
 13 .#....#.#..#+.##....###..#.#..#....#....#.#...#......##...#....
 14 ..###..#.#...+....#.....#.......###.#.#..#..#...###.....#......
 15 .#....#.#..##.+#....###..#.#..#....#....#.#...#......##...#....
 16 .#....#.#..##.#+....###..#.#..#....#....#.#...#......##...#....
 17 #.........#.....+.........#.#........#..............#.......#.#
 18 .....#...........+.#...#.....#.#.......#...#.#.#.......#.#...#.
 19 ..###..#.#...#....+.....#.......###.#.#..#..#...###.....#......
 20 .....#...........#.+...#.....#.#.......#...#.#.#.......#.#...#.
 21 .#....#.#..##.##....+##..#.#..#....#....#.#...#......##...#....
 22 .#....#.#..##.##....#+#..#.#..#....#....#.#...#......##...#....
 23 .#....#.#..##.##....##+..#.#..#....#....#.#...#......##...#....
 24 .....#...........#.#...+.....#.#.......#...#.#.#.......#.#...#.
 25 ..###..#.#...#....#.....+.......###.#.#..#..#...###.....#......
 26 .#....#.#..##.##....###..+.#..#....#....#.#...#......##...#....
 27 #.........#.....#.........+.#........#..............#.......#.#
 28 .#....#.#..##.##....###..#.+..#....#....#.#...#......##...#....
 29 #.........#.....#.........#.+........#..............#.......#.#
 30 .....#...........#.#...#.....+.#.......#...#.#.#.......#.#...#.
 31 .#....#.#..##.##....###..#.#..+....#....#.#...#......##...#....
 32 .....#...........#.#...#.....#.+.......#...#.#.#.......#.#...#.
 33 ..###..#.#...#....#.....#.......+##.#.#..#..#...###.....#......
 34 ..###..#.#...#....#.....#.......#+#.#.#..#..#...###.....#......
```

```
 35  ..###..#.#...#....#.....#.......##+.#.#..#..#...###.....#......
 36  .#....#.#..##.##....###..#.#..#....+....#.#...#......##...#....
 37  ..###..#.#...#....#.....#.......###.+.#..#..#...###.....#......
 38  #.........#.....#.........#.#........+..............#.......#.#
 39  ..###..#.#...#....#.....#.......###.#.+..#..#...###.....#......
 40  .....#...........#.#...#.....#.#.......+...#.#.#.......#.#...#.
 41  .#....#.#..##.##....###..#.#..#....#....+.#...#......##...#....
 42  ..###..#.#...#....#.....#.......###.#.#..+..#...###.....#......
 43  .#....#.#..##.##....###..#.#..#....#....#.+...#......##...#....
 44  .....#...........#.#...#.....#.#.......#...+.#.#.......#.#...#.
 45  ..###..#.#...#....#.....#.......###.#.#..#..+...###.....#......
 46  .....#...........#.#...#.....#.#.......#...#.+.#.......#.#...#.
 47  .#....#.#..##.##....###..#.#..#....#....#.#...+......##...#....
 48  .....#...........#.#...#.....#.#.......#...#.#.+.......#.#...#.
 49  ..###..#.#...#....#.....#.......###.#.#..#..#...+##.....#......
 50  ..###..#.#...#....#.....#.......###.#.#..#..#...#+#.....#......
 51  ..###..#.#...#....#.....#.......###.#.#..#..#...##+.....#......
 52  ...................................................+.......#...
 53  #.........#.....#.........#.#........#..............+.......#.#
 54  .#....#.#..##.##....###..#.#..#....#....#.#...#......+#...#....
 55  .#....#.#..##.##....###..#.#..#....#....#.#...#......#+...#....
 56  .....#...........#.#...#.....#.#.......#...#.#.#.......+.#...#.
 57  ..###..#.#...#....#.....#.......###.#.#..#..#...###.....+......
 58  .....#...........#.#...#.....#.#.......#...#.#.#.......#.+...#.
 59  .#....#.#..##.##....###..#.#..#....#....#.#...#......##...+....
 60  ...................................................#.......+...
 61  #.........#.....#.........#.#........#..............#.......+.#
 62  .....#...........#.#...#.....#.#.......#...#.#.#.......#.#...+.
 63  #.........#.....#.........#.#........#..............#.......#.+

Recurrence rate: 952/3906 (24.4%)

--- Recurrence rate null model ---
Theoretical expected recurrence rate (sum of p_i^2): 25.6%

King Wen recurrence rate:       24.4%
Mean random recurrence rate:    23.3%
King Wen percentile vs random:  72.3%
```

DNA codon mapping

```
---
DNA codon mapping
Both the I Ching (64 hexagrams) and the genetic code (64 codons) are systems
of 64 elements built from binary-like pairs: yin/yang for hexagrams, and the
base pairs A-T and G-C for DNA. Each codon is a sequence of 3 bases from the
set {A, C, G, U}, giving 4^3 = 64 possibilities. While this is a numerical
coincidence (both are 2^6 = 64), comparing their structural properties is
instructive about combinatorial systems in general.
Caveat: the mapping of bit pairs to DNA bases (00=U, 01=C, 10=A, 11=G) is one
of 24 possible assignments. Different mappings produce different codon assignments
and different degeneracy statistics. No mapping is privileged by either system.
This analysis is illustrative, not evidence of a biological connection.
---
Pos | Hex | Binary | Codon | Amino Acid | Name
----|-----|--------|-------|------------|----
01  | ䷀   | 111111 | GGG   | Gly        | The Creative
02  | ䷁   | 000000 | UUU   | Phe        | The Receptive
03  | ䷂   | 010001 | CUC   | Leu        | Difficulty at the Beginning
04  | ䷃   | 100010 | AUA   | Ile        | Youthful Folly
05  | ䷄   | 010111 | CCG   | Pro        | Waiting
06  | ䷅   | 111010 | GAA   | Glu        | Conflict
07  | ䷆   | 000010 | UUA   | Leu        | The Army
08  | ䷇   | 010000 | CUU   | Leu        | Holding Together
09  | ䷈   | 110111 | GCG   | Ala        | Small Taming
10  | ䷉   | 111011 | GAG   | Glu        | Treading
11  | ䷊   | 000111 | UCG   | Ser        | Peace
12  | ䷋   | 111000 | GAU   | Asp        | Standstill
13  | ䷌   | 111101 | GGC   | Gly        | Fellowship
14  | ䷍   | 101111 | AGG   | Arg        | Great Possession
15  | ䷎   | 000100 | UCU   | Ser        | Modesty
```

```
16  | ䷏  | 001000 | UAU  | Tyr        | Enthusiasm
17  | ䷐  | 011001 | CAC  | His        | Following
18  | ䷑  | 100110 | ACA  | Thr        | Work on the Decayed
19  | ䷒  | 000011 | UUG  | Leu        | Approach
20  | ䷓  | 110000 | GUU  | Val        | Contemplation
21  | ䷔  | 101001 | AAC  | Asn        | Biting Through
22  | ䷕  | 100101 | ACC  | Thr        | Grace
23  | ䷖  | 100000 | AUU  | Ile        | Splitting Apart
24  | ䷗  | 000001 | UUC  | Phe        | Return
25  | ䷘  | 111001 | GAC  | Asp        | Innocence
26  | ䷙  | 100111 | ACG  | Thr        | Great Taming
27  | ䷚  | 100001 | AUC  | Ile        | Nourishment
28  | ䷛  | 011110 | CGA  | Arg        | Great Preponderance
29  | ䷜  | 010010 | CUA  | Leu        | The Abysmal
30  | ䷝  | 101101 | AGC  | Ser        | The Clinging
31  | ䷞  | 011100 | CGU  | Arg        | Influence
32  | ䷟  | 001110 | UGA  | Stop       | Duration
33  | ䷠  | 111100 | GGU  | Gly        | Retreat
34  | ䷡  | 001111 | UGG  | Trp        | Great Power
35  | ䷢  | 101000 | AAU  | Asn        | Progress
36  | ䷣  | 000101 | UCC  | Ser        | Darkening of the Light
37  | ䷤  | 110101 | GCC  | Ala        | The Family
38  | ䷥  | 101011 | AAG  | Lys        | Opposition
39  | ䷦  | 010100 | CCU  | Pro        | Obstruction
40  | ䷧  | 001010 | UAA  | Stop       | Deliverance
41  | ䷨  | 100011 | AUG  | Met        | Decrease
42  | ䷩  | 110001 | GUC  | Val        | Increase
43  | ䷪  | 011111 | CGG  | Arg        | Breakthrough
44  | ䷫  | 111110 | GGA  | Gly        | Coming to Meet
45  | ䷬  | 011000 | CAU  | His        | Gathering Together
46  | ䷭  | 000110 | UCA  | Ser        | Pushing Upward
47  | ䷮  | 011010 | CAA  | Gln        | Oppression
48  | ䷯  | 010110 | CCA  | Pro        | The Well
49  | ䷰  | 011101 | CGC  | Arg        | Revolution
50  | ䷱  | 101110 | AGA  | Arg        | The Cauldron
51  | ䷲  | 001001 | UAC  | Tyr        | The Arousing
52  | ䷳  | 100100 | ACU  | Thr        | Keeping Still
53  | ䷴  | 110100 | GCU  | Ala        | Development
54  | ䷵  | 001011 | UAG  | Stop       | The Marrying Maiden
55  | ䷶  | 001101 | UGC  | Cys        | Abundance
56  | ䷷  | 101100 | AGU  | Ser        | The Wanderer
57  | ䷸  | 110110 | GCA  | Ala        | The Gentle
58  | ䷹  | 011011 | CAG  | Gln        | The Joyous
59  | ䷺  | 110010 | GUA  | Val        | Dispersion
60  | ䷻  | 010011 | CUG  | Leu        | Limitation
61  | ䷼  | 110011 | GUG  | Val        | Inner Truth
62  | ䷽  | 001100 | UGU  | Cys        | Small Preponderance
63  | ䷾  | 010101 | CCC  | Pro        | After Completion
64  | ䷿  | 101010 | AAA  | Lys        | Before Completion

Unique amino acids in King Wen order: 21
Total possible amino acids + stop: 21

--- Degeneracy comparison ---
In DNA, many single-base changes preserve the amino acid (the code is 'degenerate').
Do single-line hexagram changes preserve the mapped amino acid?
Single-line changes that preserve amino acid: 100/384 (26.0%)
```

Shannon entropy analysis

```
---
Shannon entropy analysis
Entropy measures disorder: high entropy means the difference values are spread
evenly across all possibilities (random-looking), low entropy means certain values
dominate (structured). We compare the King Wen wave's entropy against thousands
of random permutations of the same 64 hexagrams. If King Wen's entropy is
unusually low, the sequence is more structured than random chance would produce.
---
King Wen difference wave entropy: 2.0759 bits
Maximum entropy (all 7 values): 2.8074 bits
Maximum entropy (5 observed values): 2.3219 bits
```

```
Mean entropy of random permutations: 2.1934 bits
Min random entropy observed: 1.7320 bits
Max random entropy observed: 2.4964 bits
King Wen percentile: 12.8% (lower = more structured)
Effect size (Cohen's d): -1.14 (negative = more structured than random)

--- Entropy conditioned on pair constraint ---
The unconstrained comparison above may be misleading: the pair structure
itself constrains the entropy. How does King Wen compare against random
orderings that also satisfy the pair constraint?
Mean pair-constrained entropy: 2.2341 bits
King Wen percentile (pair-constrained): 5.9%
(Similar to unconstrained percentile of 12.8%.)

--- Distribution comparison ---
Value    King Wen Expected (random avg)
  0            0                   0.0
  1            2                   5.8
  2           20                  15.0
  3           13                  20.2
  4           19                  15.0
  5            0                   6.0
  6            9                   1.0
```

Path analysis (graph theory)

```
---
Path analysis (graph theory)
Imagine the 64 hexagrams as cities on a map, where the 'distance' between any
two is the number of lines that differ. The King Wen sequence is a route that
visits all 64 cities. Is it a short, efficient route (nearby hexagrams placed
next to each other), or a long, wandering one? We compare its total distance
against random routes and a greedy 'always go to the nearest unvisited city'
algorithm. This reveals whether the King Wen ordering was optimized for smooth
transitions or had other priorities.
---
King Wen total path length:  211 (sum of all line changes)
Greedy nearest-neighbor:     75
Mean random path length:     192.1
Min random observed:         155
Max random observed:         228
King Wen percentile:         97.2% (lower = shorter path)
Effect size (Cohen's d):     +2.00

--- Theoretical bounds ---
Minimum possible (63 transitions of 1): 63
Maximum possible (63 transitions of 6): 378
King Wen uses 55.8% of maximum path length

--- Pair-constrained comparison ---
The above compares against fully random orderings. A fairer comparison is
against random orderings that also preserve the pair structure (each pair
of hexagrams stays adjacent). This is the right null model for asking
whether King Wen's path length is unusual GIVEN its pair constraint.
Mean pair-constrained path length: 214.2
Min pair-constrained observed:     193
Max pair-constrained observed:     239
King Wen percentile (pair-constrained): 28.3%
Effect size (Cohen's d, pair-constrained): -0.49
```

Constraint satisfaction analysis

```
---
Constraint satisfaction analysis
The King Wen sequence satisfies several constraints simultaneously:
  1. All 32 consecutive pairs are either reverse or inverse (never unrelated)
  2. No consecutive hexagrams differ by exactly 5 lines
  3. No consecutive hexagrams are identical (0-line transitions)
```

```
How rare is it for a random ordering to satisfy all these at once? We test
random permutations against each constraint individually and combined.
---
Results from 10,000 random permutations:
  All pairs reverse/inverse:       0 (0.000%)
  No 5-line transitions:          22 (0.22%)
  Both constraints together:       0 (0.0000%)
  No random permutation satisfied both constraints.
  Statistical note: 0/10,000 gives a 95% upper bound of <0.0300%
  (i.e., the true rate is less than ~1 in 3,333, but we cannot
  distinguish 'impossible' from 'extremely rare' with this sample size).

--- Conditional probability: P(no-5 | pair constraint) ---
The pair structure constrains transitions within pairs (always even or 6),
so 5-line transitions can only occur at the 31 between-pair boundaries.
How often do pair-constrained orderings also avoid 5-line transitions?
  Pair-constrained trials: 100,000
  Also satisfy no-5:       4,274 (4.27%)
  Approximately 1 in 23 pair-constrained orderings avoid 5-line transitions.
  The no-5 property is uncommon but not extraordinary among pair-constrained orderings.

--- Sensitivity analysis: reversed bit convention ---
What if bit 0 = top line instead of bottom? All binary values reverse,
changing pair types and the difference wave. Key properties tested:
  Pair structure preserved:  Yes
  No-5 property preserved:   Yes
  Difference wave identical: Yes
  Original entropy: 2.0759, reversed: 2.0759
  Original path length: 211, reversed: 211
  All key properties are invariant under bit reversal.
```

Bootstrap confidence intervals

```
---
Bootstrap confidence intervals
The Monte Carlo results (e.g., '0.18% of random orderings avoid 5-line
transitions') are estimates based on sampling. Bootstrap resampling quantifies
how precise those estimates are by repeatedly resampling from the results and
computing confidence intervals. Narrower intervals = more reliable estimates.
---
Base trials: 10,000
Bootstrap resamples: 1000

No-5-line-transition rate: 0.170%
95% confidence interval: [0.100%, 0.260%]
Interval width: 0.160 percentage points

Approximately 1 in 588 random orderings
95% CI: 1 in 385 to 1 in 1000

Note: These CIs measure the precision of the Monte Carlo estimate, not
fundamental uncertainty about the true proportion. They reflect how much
the estimate would vary if you re-ran the simulation with 10,000 trials.
Increasing --trials narrows these CIs because the estimate becomes more precise.
```

Monte Carlo analysis

```
---
Monte Carlo analysis (10,000 random permutations)
The King Wen sequence has a striking property: no two consecutive hexagrams
differ by exactly 5 lines. With 6 lines per hexagram and 7 possible difference
values (0-6), is avoiding 5 remarkable or just a coincidence? To find out, we
randomly shuffle the 64 hexagrams thousands of times and check how often a
random ordering also avoids 5-line transitions. The rarer it is, the less
plausible chance becomes as an explanation for the avoidance.
---
Permutations with no 5-line transitions: 21/10,000 (0.21%)
Approximately 1 in 476 random orderings share this property.
```

```
Odds ratio against random: 475:1
```